

LAB2 : Cassandra

Perform the following DB operations using Cassandra.

- Create a keyspace by name Employee
- Create a column family by name Employee-Info with attributes Emp_Id Primary Key, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name
- Insert the values into the table in batch
- Update Employee name and Department of Emp-Id 121
- Sort the details of Employee records based on salary
- Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
- Update the altered table to add project names.
- Create a TTL of 15 seconds to display the values of Employees.

```
bmscscse@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~  
bmscscse@bmscscse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~$ cqlsh  
Connected to Test Cluster at 127.0.0.1:9042  
[cqlsh 6.1.0 | Cassandra 4.1.8 | CQL spec 3.4.6 | Native protocol v5]  
Use HELP for help.  
cqlsh> create keyspace emp with replication = {'class':'SimpleStrategy','replication_factor':1};  
cqlsh> describe keyspaces;  
  
emp          library1   students    system_distributed  system_views  
employee     library2   system      system_schema       system_virtual_schema  
library       s1        system_auth system_traces  
  
cqlsh> create table empt(id int primary key, name text, dept text, date timestamp, salary double);  
InvalidRequest: Error from server: code=2200 [Invalid query] message="No keyspace has been specified. USE a keyspace, or explicitly specify keyspace.tablename"  
cqlsh> use emp  
...  
cqlsh> use emp;  
cqlsh:emp> create table empt(id int primary key, name text, dept text, date timestamp, salary double);  
cqlsh:emp> describe tables;  
  
empt  
  
cqlsh:emp> begin batch  
... insert into empt(id,name,dept,date,salary)  
... values(1,'a','xx','2026-01-01',10000);  
... insert into empt(id,name,dept,date,salary)  
... values(2,'b','yy','2026-01-02',20000);  
... insert into empt(id,name,dept,date,salary)  
... values(3,'c','zz','2026-01-03',30000);  
... apply batch;  
cqlsh:emp> select * from empt;  
  
id | date                    | dept | name | salary  
---+-----  
1 | 2025-12-31 18:30:00.000000+0000 | xx   | a    | 10000  
2 | 2026-01-01 18:30:00.000000+0000 | yy   | b    | 20000  
3 | 2026-01-02 18:30:00.000000+0000 | zz   | c    | 30000  
  
(3 rows)  
cqlsh:emp> insert into empt(id,name,dept,date,salary) values(0,'p','pp','2026-01-04',40000);  
cqlsh:emp> select * from empt;  
  
id | date                    | dept | name | salary  
---+-----  
1 | 2025-12-31 18:30:00.000000+0000 | xx   | a    | 10000  
0 | 2026-01-03 18:30:00.000000+0000 | pp   | p    | 40000  
2 | 2026-01-01 18:30:00.000000+0000 | yy   | b    | 20000  
3 | 2026-01-02 18:30:00.000000+0000 | zz   | c    | 30000  
  
(4 rows)  
cqlsh:emp> update empt set dept='xx',name='a',salary=10000 where id=0;  
cqlsh:emp> select * from empt;
```

(3 rows)

```
cqlsh:emp> insert into emp(id,name,dept,date,salary) values(0,'p','pp','2026-01-04',40000);
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

(4 rows)

```
cqlsh:emp> update emp set dept='xx',name='a',salary=10000 where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	xx	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

(4 rows)

```
cqlsh:emp> delete dept='xx' from emp where id=0;
SyntaxException: line 1:11 no viable alternative at input '=' (delete [dept]=...)
cqlsh:emp> delete dept from emp where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
0	2026-01-03 18:30:00.000000+0000	null	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

(4 rows)

```
cqlsh:emp> delete dept:'xx' from emp where id=0;
SyntaxException: line 1:11 no viable alternative at input ':' (delete [dept]:...)
cqlsh:emp> delete from emp where id=0;
cqlsh:emp> select * from emp;
```

id	date	dept	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	c	30000

(3 rows)

```
cqlsh:emp> alter table emp add grade set<text>;
cqlsh:emp> select * from emp;
```

id	date	dept	grade	name	salary
----	------	------	-------	------	--------

(3 rows)

```
cqlsh:emp> alter table emp add grade set<text>;
```

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	null	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	null	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	null	c	30000

(3 rows)

```
cqlsh:emp> update emp set grade={'a','b'} where id=1;
```

```
cqlsh:emp> update emp set grade={'a','b'} where id=2;
```

```
cqlsh:emp> update emp set grade={'a','b','c'} where id=3;
```

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	c	30000

(3 rows)

```
cqlsh:emp> alter table emp add marks list<int>;
```

```
cqlsh:emp> update emp set list={1,2,3} where id=0;
```

```
InvalidRequest: Error from server: code=2200 [Invalid query] message="Undefined column name list"
```

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	null	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	null	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	null	c	30000

(3 rows)

```
cqlsh:emp> update emp set marks={1,2,3} where id=1;
```

```
InvalidRequest: Error from server: code=2200 [Invalid query] message="Invalid set literal for marks"
```

```
cqlsh:emp> update emp set marks=[1,2,3] where id=1;
```

```
cqlsh:emp> update emp set marks=[1,2,3] where id=2;
```

```
cqlsh:emp> update emp set marks=[1] where id=3;
```

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(3 rows)

```
cqlsh:emp> select dept as "department" from emp;
```

```
department
```

```
cqlsh:emp> update emp set marks=[1,2,3] where id=2;
cqlsh:emp> update emp set marks=[1] where id=3;
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(3 rows)

```
cqlsh:emp> select dept as "department" from emp;
```

department
xx
yy
zz

(3 rows)

```
cqlsh:emp> select dept as 'department' from emp;
```

SyntaxException: line 1:15 no viable alternative at input 'department' (select dept as ['depart

```
cqlsh:emp> insert into emp(id,name,dept,date,salary) values(0,'p','pp','2026-01-04',40000) using
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	null	null	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(4 rows)

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
0	2026-01-03 18:30:00.000000+0000	pp	null	null	p	40000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(4 rows)

```
cqlsh:emp> select * from emp;
```

id	date	dept	grade	marks	name	salary
1	2025-12-31 18:30:00.000000+0000	xx	{ 'a', 'b' }	[1, 2, 3]	a	10000
2	2026-01-01 18:30:00.000000+0000	yy	{ 'a', 'b' }	[1, 2, 3]	b	20000
3	2026-01-02 18:30:00.000000+0000	zz	{ 'a', 'b', 'c' }	[1]	c	30000

(3 rows)

```
cqlsh:emp>
```